NAME-: Devanshish Ganatra

SUBJECT-: Computer Network

ROLL NO-: 71

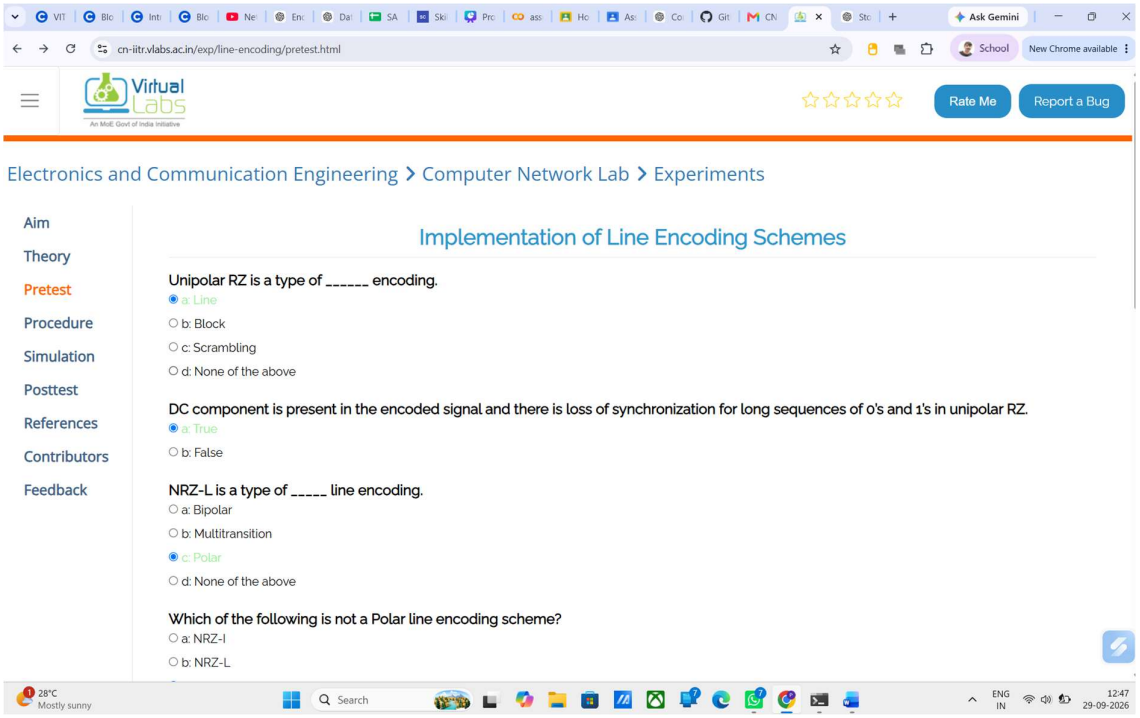
BATCH NO-: B3

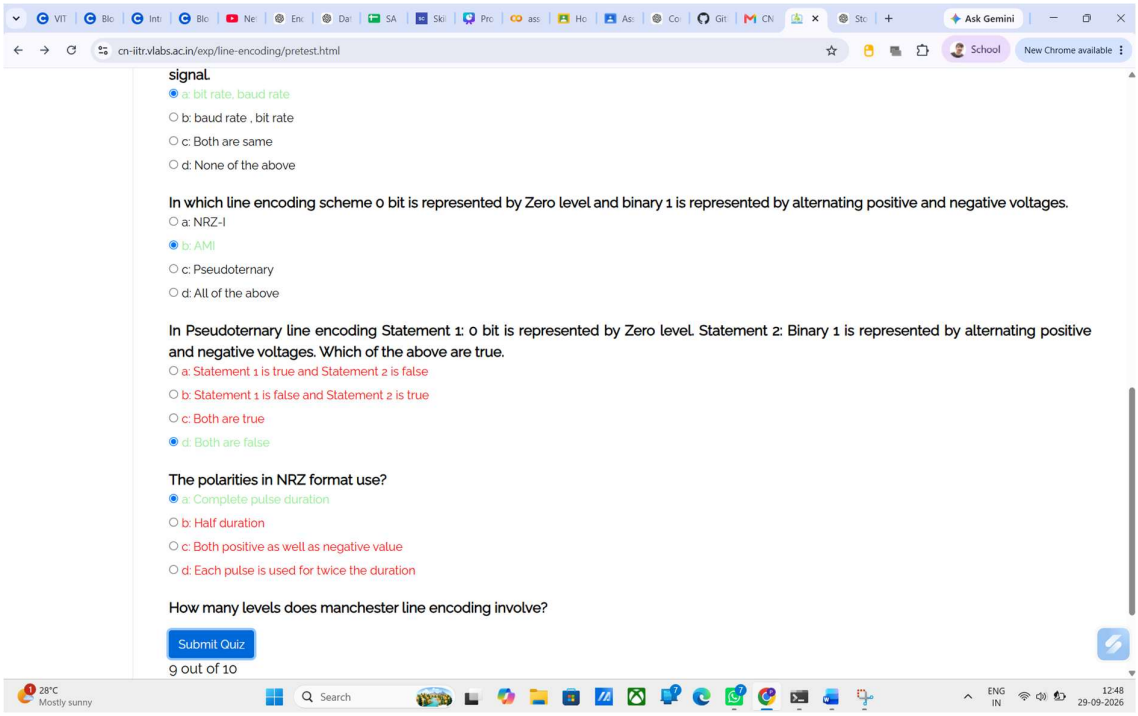
PRN-: 12415046

Virtual Lab

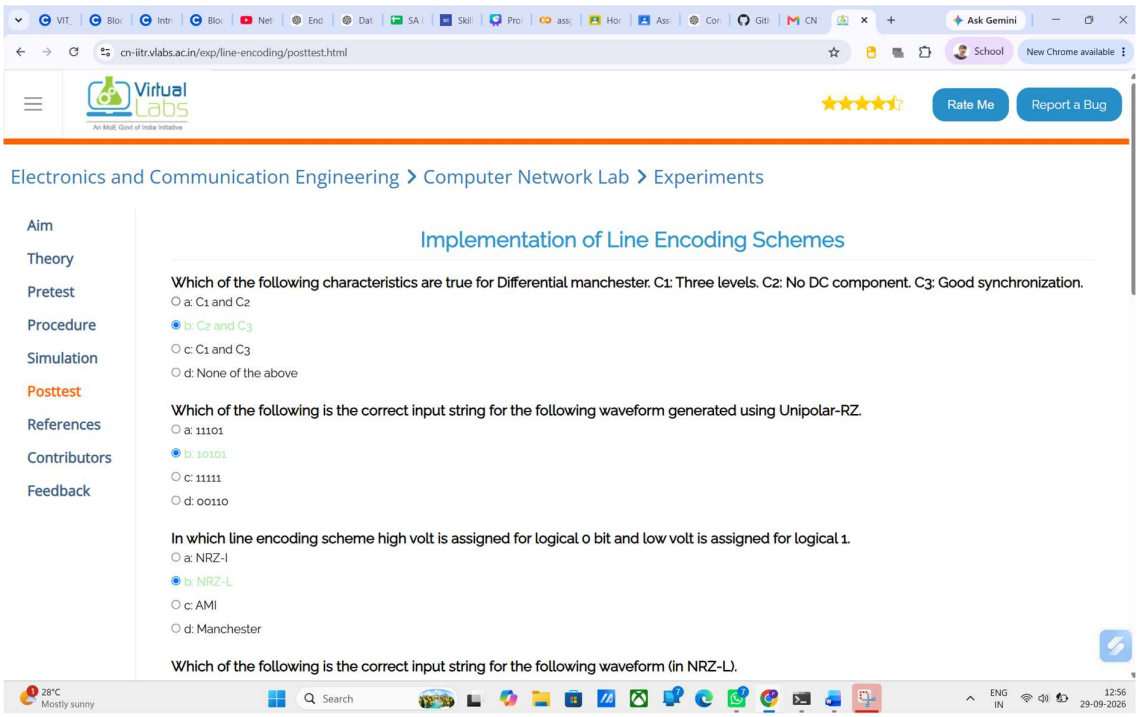
Title-: **Implementation of Line Encoding Schemes**

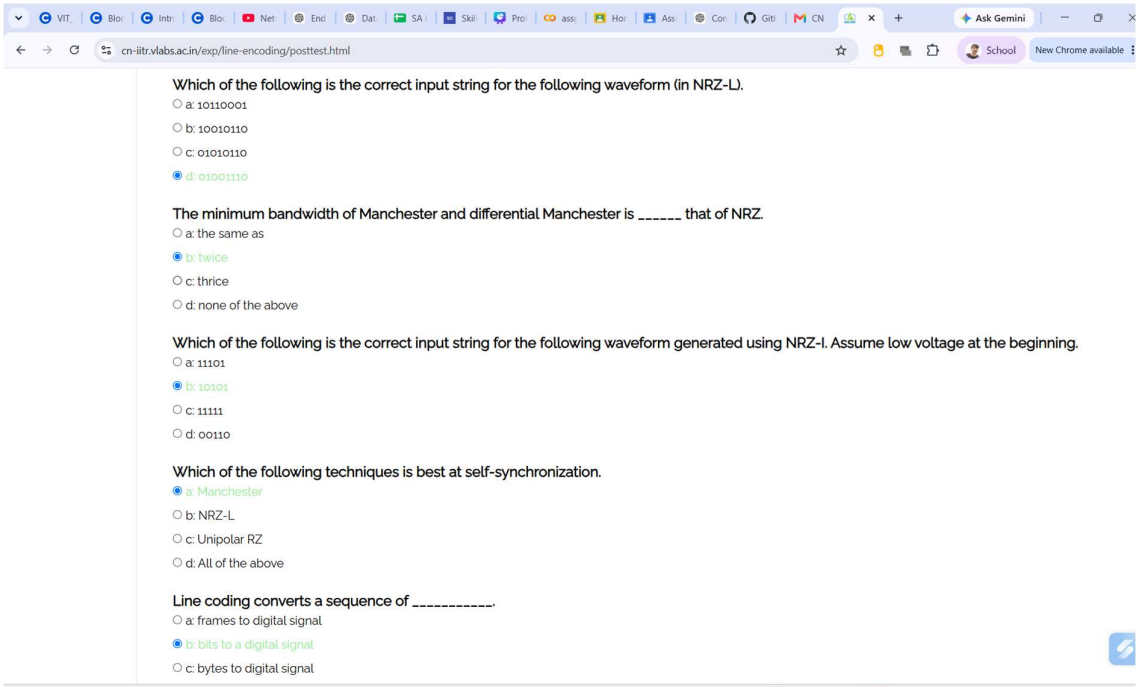
Pretest-:




cn-iitr.vlabs.ac.in/exp/line-encoding/pretest.html
signal.
a: bit rate, baud rate
b: baud rate , bit rate
c: Both are same
d: None of the above
In which line encoding scheme 0 bit is represented by Zero level and binary 1 is represented by alternating positive and negative voltages.
a: NRZ-I
b: AMI
c: Pseudoternary
d: All of the above
In Pseudoternary line encoding Statement 1: 0 bit is represented by Zero level. Statement 2: Binary 1 is represented by alternating positive and negative voltages. Which of the above are true.
a: Statement 1 is true and Statement 2 is false
b: Statement 1 is false and Statement 2 is true
c: Both are true
d: Both are false
The polarities in NRZ format use?
a: Complete pulse duration
b: Half duration
c: Both positive as well as negative value
d: Each pulse is used for twice the duration
How many levels does manchester line encoding involve?
Submit Quiz
9 out of 10

Posttest-:


cn-iitr.vlabs.ac.in/exp/line-encoding/posttest.html
Virtual Labs
Rate Me
Report a Bug
Electronics and Communication Engineering > Computer Network Lab > Experiments
Aim
Theory
Pretest
Procedure
Simulation
Posttest
References
Contributors
Feedback
Implementation of Line Encoding Schemes
Which of the following characteristics are true for Differential manchester. C1: Three levels. C2: No DC component. C3: Good synchronization.
a: C1 and C2
b: C2 and C3
c: C1 and C3
d: None of the above
Which of the following is the correct input string for the following waveform generated using Unipolar-RZ.
a: 11101
b: 10101
c: 11111
d: 00110
In which line encoding scheme high volt is assigned for logical 0 bit and low volt is assigned for logical 1.
a: NRZ-I
b: NRZ-L
c: AMI
d: Manchester
Which of the following is the correct input string for the following waveform (in NRZ-L).



Which of the following is the correct input string for the following waveform (in NRZ-L).

- a: 10110001
- b: 10010110
- c: 01010110
- d: 01001110

The minimum bandwidth of Manchester and differential Manchester is _____ that of NRZ.

- a: the same as
- b: twice
- c: thrice
- d: none of the above

Which of the following is the correct input string for the following waveform generated using NRZ-I. Assume low voltage at the beginning.

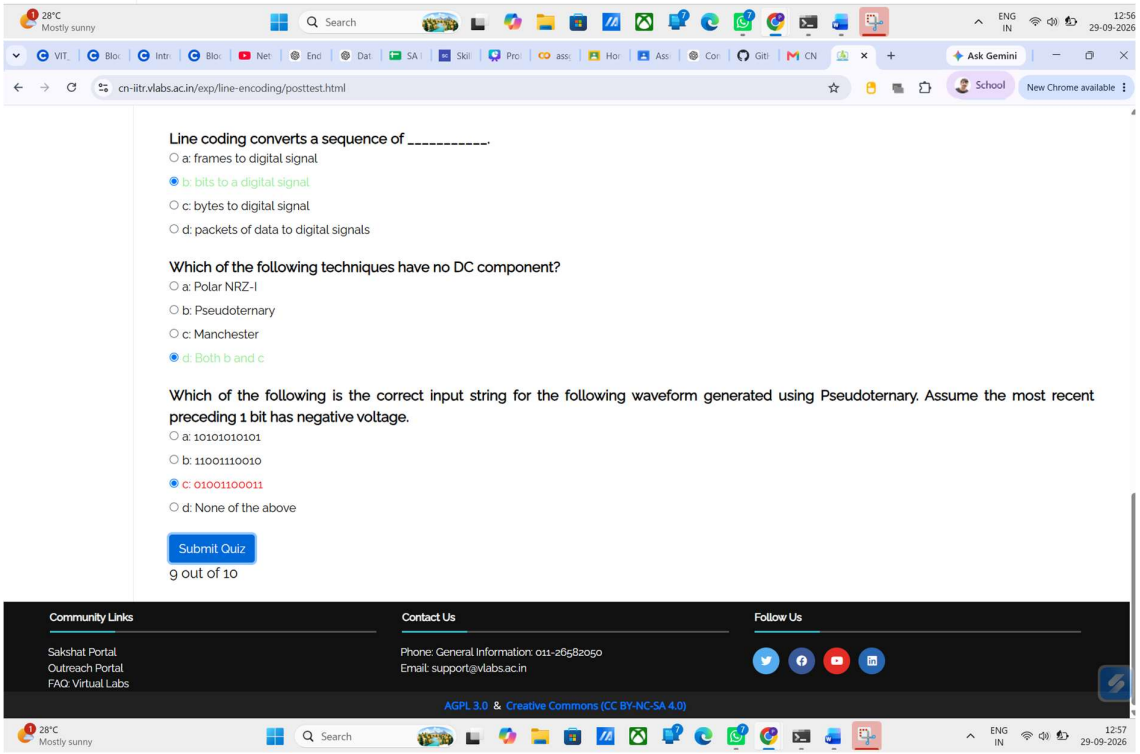
- a: 11101
- b: 10101
- c: 11111
- d: 00110

Which of the following techniques is best at self-synchronization.

- a: Manchester
- b: NRZ-L
- c: Unipolar RZ
- d: All of the above

Line coding converts a sequence of _____.

- a: frames to digital signal
- b: bits to a digital signal
- c: bytes to digital signal



Line coding converts a sequence of _____.

- a: frames to digital signal
- b: bits to a digital signal
- c: bytes to digital signal
- d: packets of data to digital signals

Which of the following techniques have no DC component?

- a: Polar NRZ-I
- b: Pseudoternary
- c: Manchester
- d: Both b and c

Which of the following is the correct input string for the following waveform generated using Pseudoternary. Assume the most recent preceding 1 bit has negative voltage.

- a: 10101010101
- b: 11001110010
- c: 01001100011
- d: None of the above

Submit Quiz

9 out of 10

Community Links

Sakshat Portal
Outreach Portal
FAQ: Virtual Labs

Contact Us

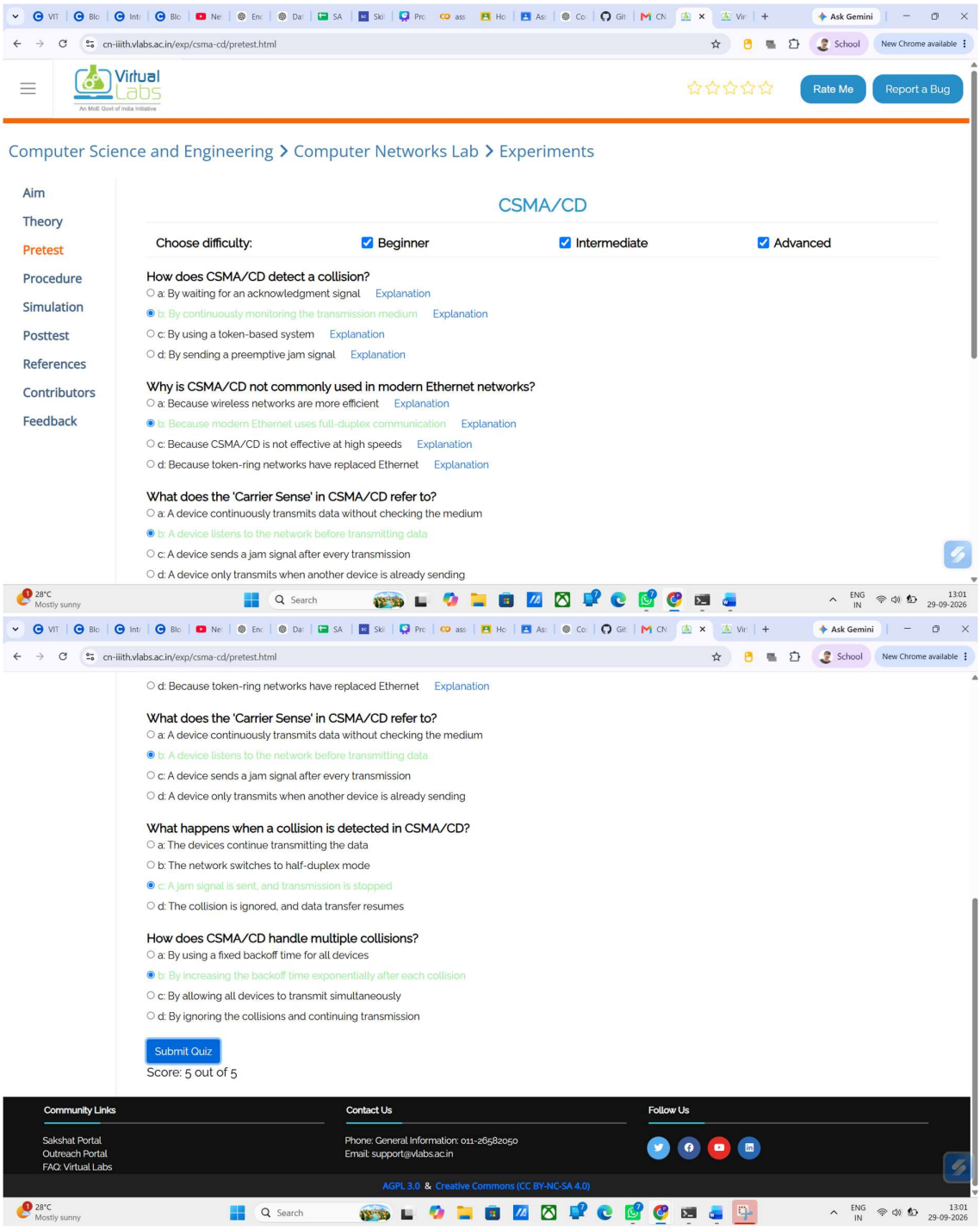
Phone: General Information: 011-26582050
Email: support@vlabs.ac.in

Follow Us

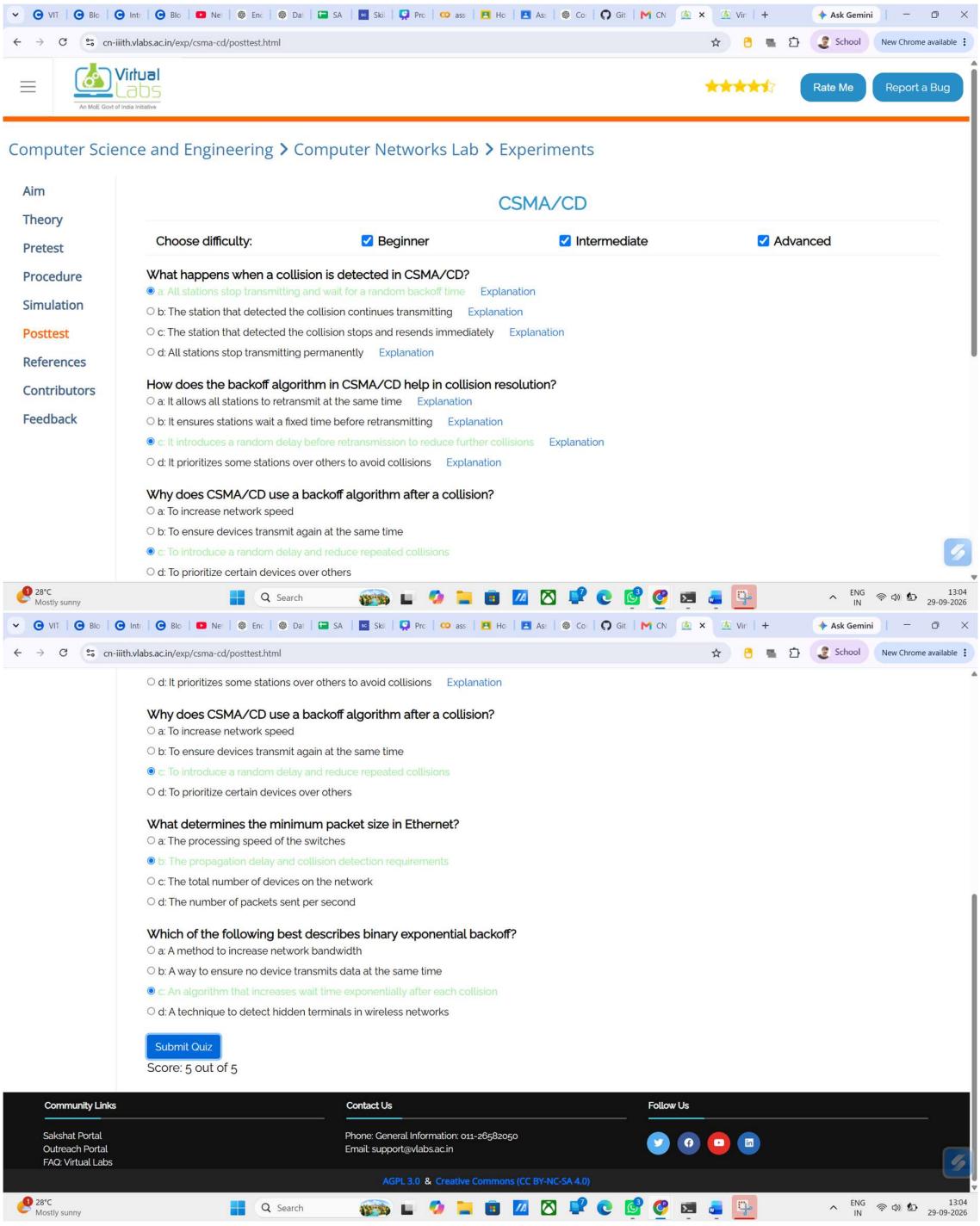
AGPL 3.0 & Creative Commons (CC BY-NC-SA 4.0)

Title-: Carrier Sense Multiple Access with Collision Detection (CSMA/CD)

Pretest-:

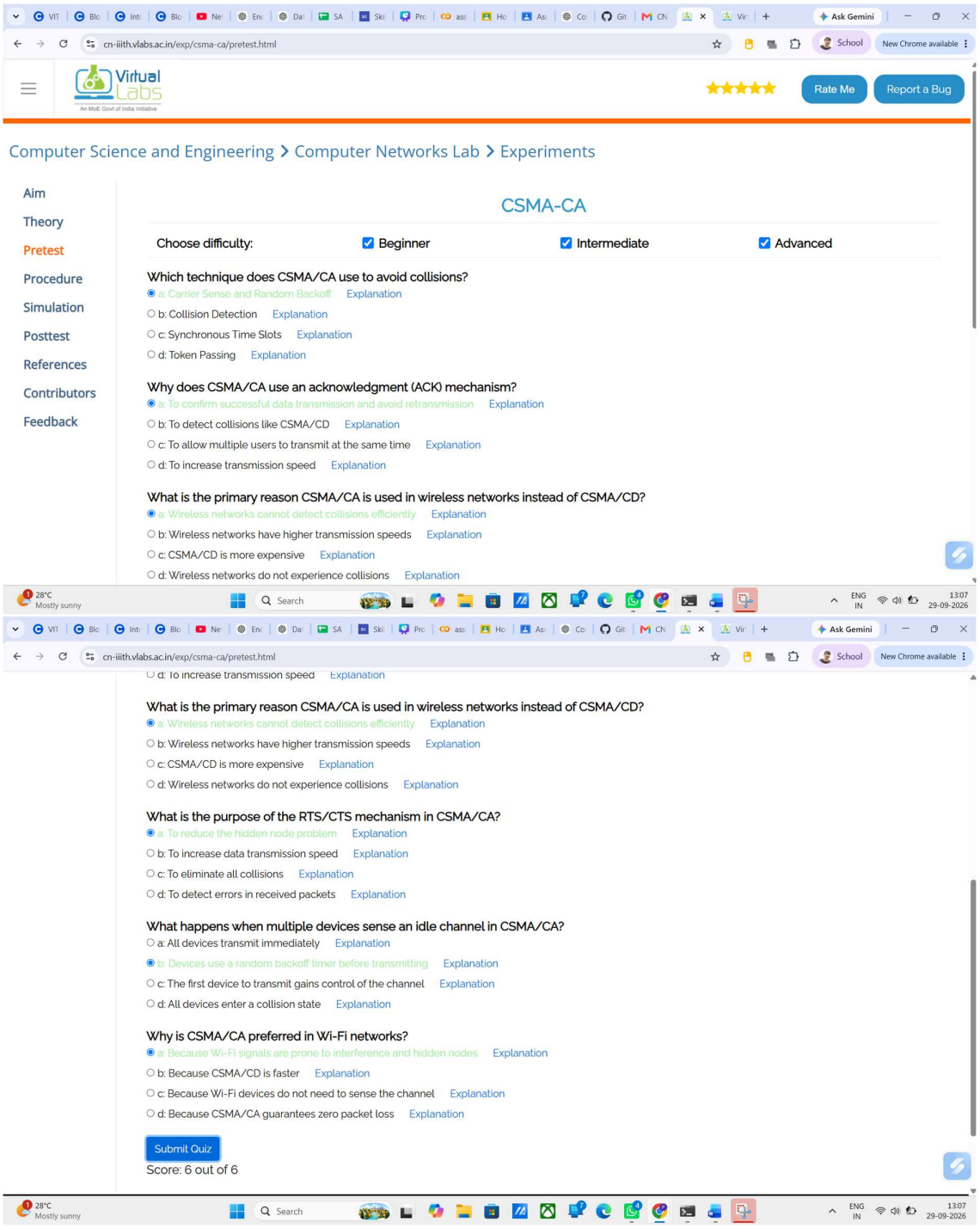


Posttest-:

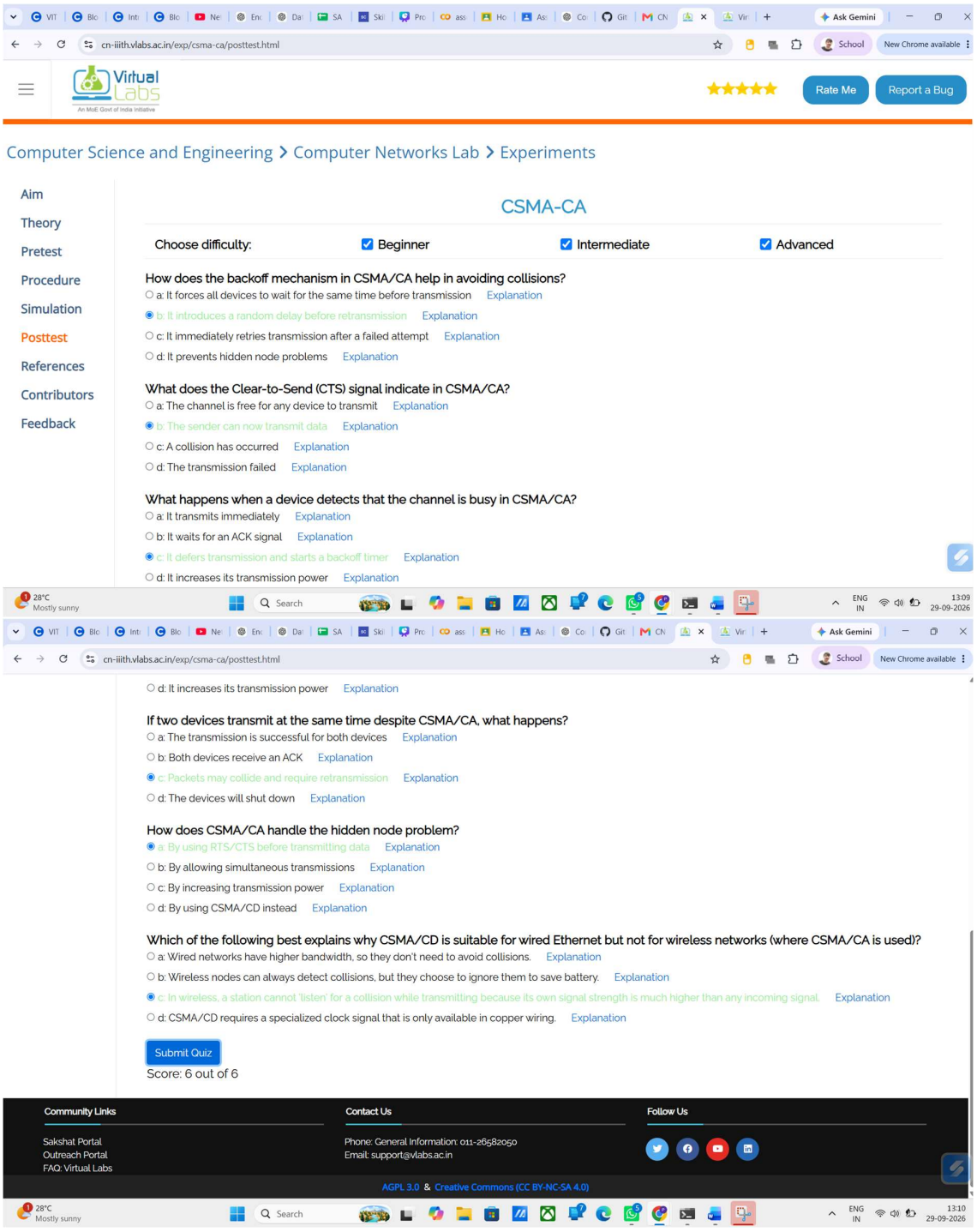

cn-iiith.vlabs.ac.in/exp/csma-cd/posttest.html
Virtual Labs
Rate Me
Report a Bug
Computer Science and Engineering > Computer Networks Lab > Experiments
Aim
Theory
Pretest
Procedure
Simulation
Posttest
References
Contributors
Feedback
CSMA/CD
Choose difficulty:
Beginner
Intermediate
Advanced
What happens when a collision is detected in CSMA/CD?
a: All stations stop transmitting and wait for a random backoff time Explanation
b: The station that detected the collision continues transmitting Explanation
c: The station that detected the collision stops and resends immediately Explanation
d: All stations stop transmitting permanently Explanation
How does the backoff algorithm in CSMA/CD help in collision resolution?
a: It allows all stations to retransmit at the same time Explanation
b: It ensures stations wait a fixed time before retransmitting Explanation
c: It introduces a random delay before retransmission to reduce further collisions Explanation
d: It prioritizes some stations over others to avoid collisions Explanation
Why does CSMA/CD use a backoff algorithm after a collision?
a: To increase network speed
b: To ensure devices transmit again at the same time
c: To introduce a random delay and reduce repeated collisions
d: To prioritize certain devices over others
What determines the minimum packet size in Ethernet?
a: The processing speed of the switches
b: The propagation delay and collision detection requirements
c: The total number of devices on the network
d: The number of packets sent per second
Which of the following best describes binary exponential backoff?
a: A method to increase network bandwidth
b: A way to ensure no device transmits data at the same time
c: An algorithm that increases wait time exponentially after each collision
d: A technique to detect hidden terminals in wireless networks
Submit Quiz
Score: 5 out of 5
Community Links
Sakshat Portal
Outreach Portal
FAQ: Virtual Labs
Contact Us
Phone: General Information: 011-26582050
Email: support@vlabs.ac.in
Follow Us
AGPL 3.0 & Creative Commons (CC BY-NC-SA 4.0)

Title-: Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)

Pretest-:

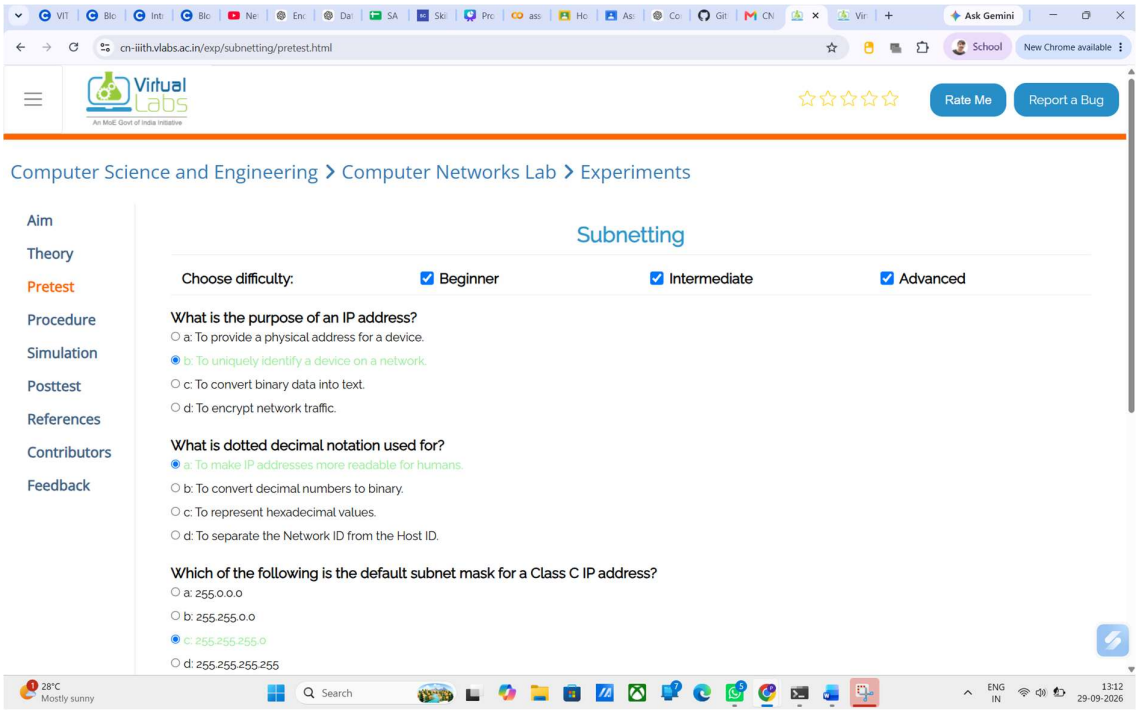


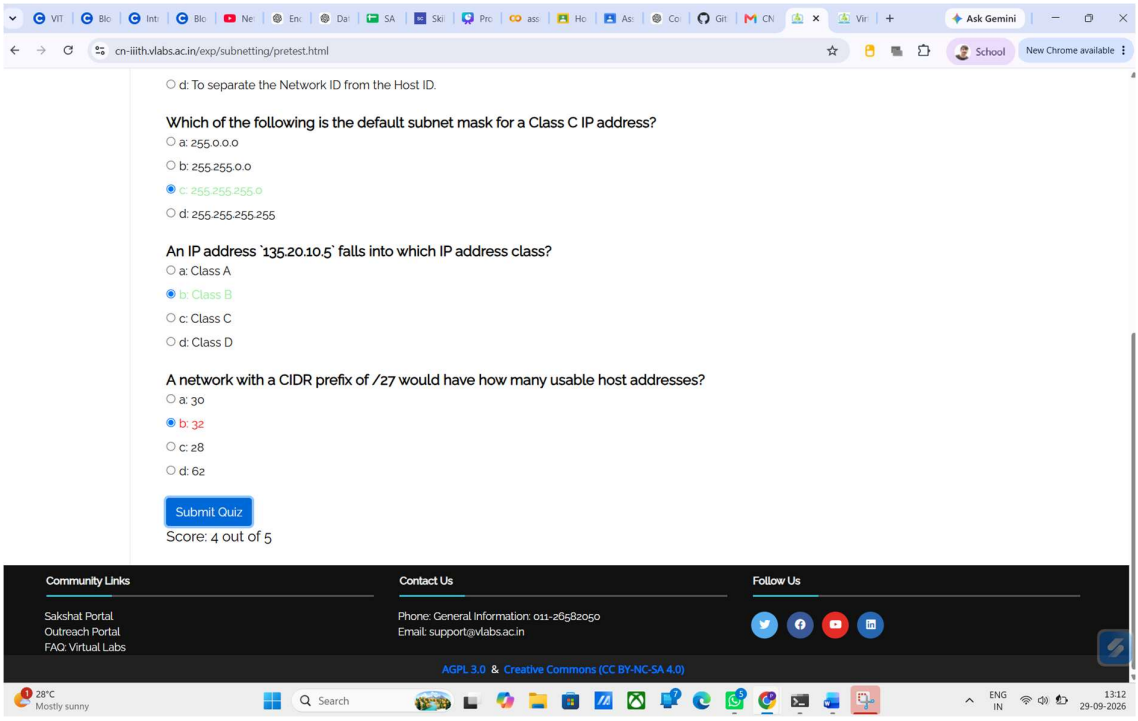
Posttest-:


cn-iiith.vlabs.ac.in/exp/csma-ca/posttest.html
Virtual Labs
Rate Me
Report a Bug
Computer Science and Engineering > Computer Networks Lab > Experiments
Aim
Theory
Pretest
Procedure
Simulation
Posttest
References
Contributors
Feedback
CSMA-CA
Choose difficulty: Beginner Intermediate Advanced
How does the backoff mechanism in CSMA/CA help in avoiding collisions?
a: It forces all devices to wait for the same time before transmission Explanation
b: It introduces a random delay before retransmission Explanation
c: It immediately retries transmission after a failed attempt Explanation
d: It prevents hidden node problems Explanation
What does the Clear-to-Send (CTS) signal indicate in CSMA/CA?
a: The channel is free for any device to transmit Explanation
b: The sender can now transmit data Explanation
c: A collision has occurred Explanation
d: The transmission failed Explanation
What happens when a device detects that the channel is busy in CSMA/CA?
a: It transmits immediately Explanation
b: It waits for an ACK signal Explanation
c: It defers transmission and starts a backoff timer Explanation
d: It increases its transmission power Explanation
d: It increases its transmission power Explanation
If two devices transmit at the same time despite CSMA/CA, what happens?
a: The transmission is successful for both devices Explanation
b: Both devices receive an ACK Explanation
c: Packets may collide and require retransmission Explanation
d: The devices will shut down Explanation
How does CSMA/CA handle the hidden node problem?
a: By using RTS/CTS before transmitting data Explanation
b: By allowing simultaneous transmissions Explanation
c: By increasing transmission power Explanation
d: By using CSMA/CD instead Explanation
Which of the following best explains why CSMA/CD is suitable for wired Ethernet but not for wireless networks (where CSMA/CA is used)?
a: Wired networks have higher bandwidth, so they don't need to avoid collisions. Explanation
b: Wireless nodes can always detect collisions, but they choose to ignore them to save battery. Explanation
c: In wireless, a station cannot 'listen' for a collision while transmitting because its own signal strength is much higher than any incoming signal. Explanation
d: CSMA/CD requires a specialized clock signal that is only available in copper wiring. Explanation
Submit Quiz
Score: 6 out of 6
Community Links
Sakshat Portal
Outreach Portal
FAQ: Virtual Labs
Contact Us
Phone: General Information: 011-26582050
Email: support@vlabs.ac.in
Follow Us
AGPL 3.0 & Creative Commons (CC BY-NC-SA 4.0)

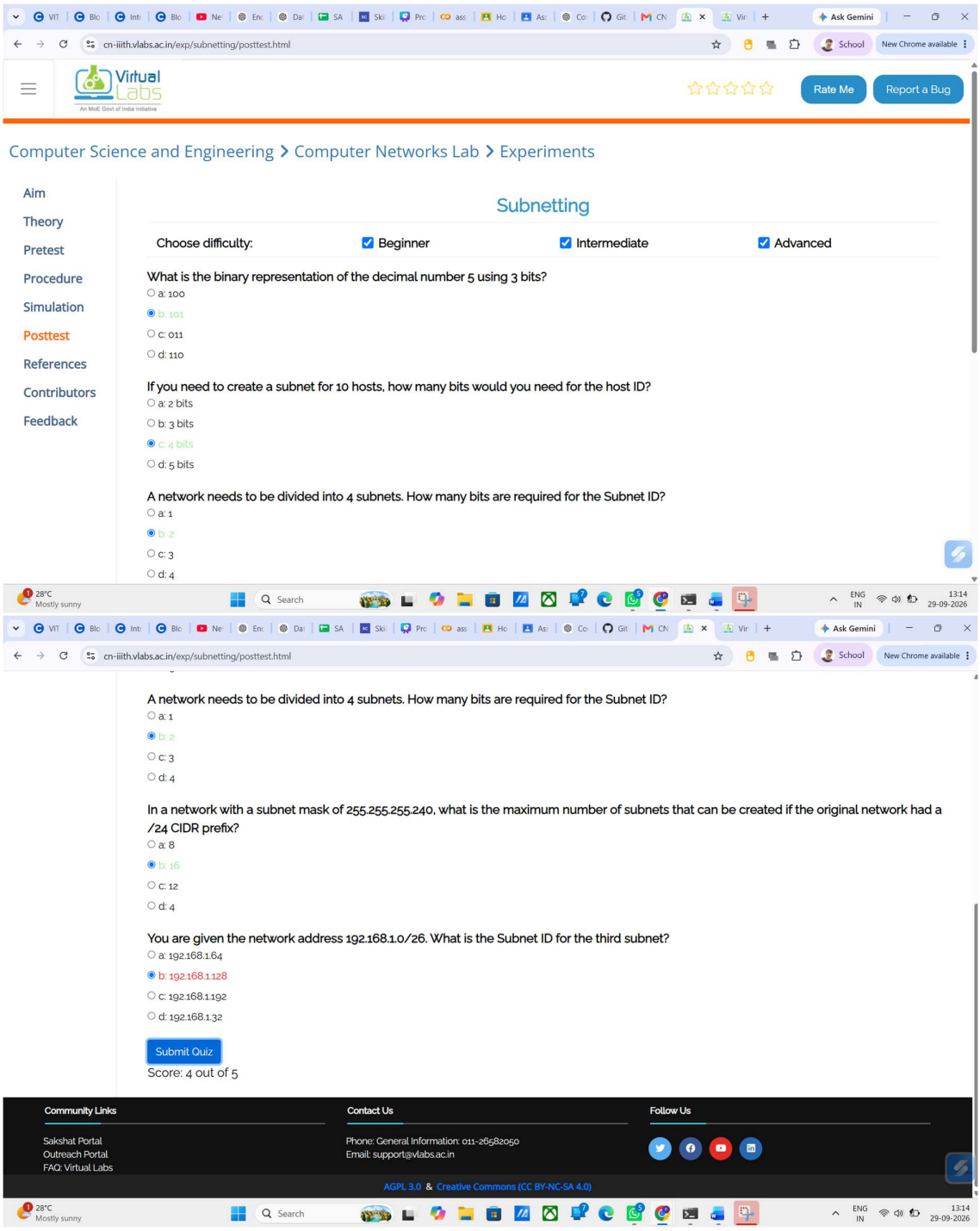
Title-:Subnetting

Pretest-:



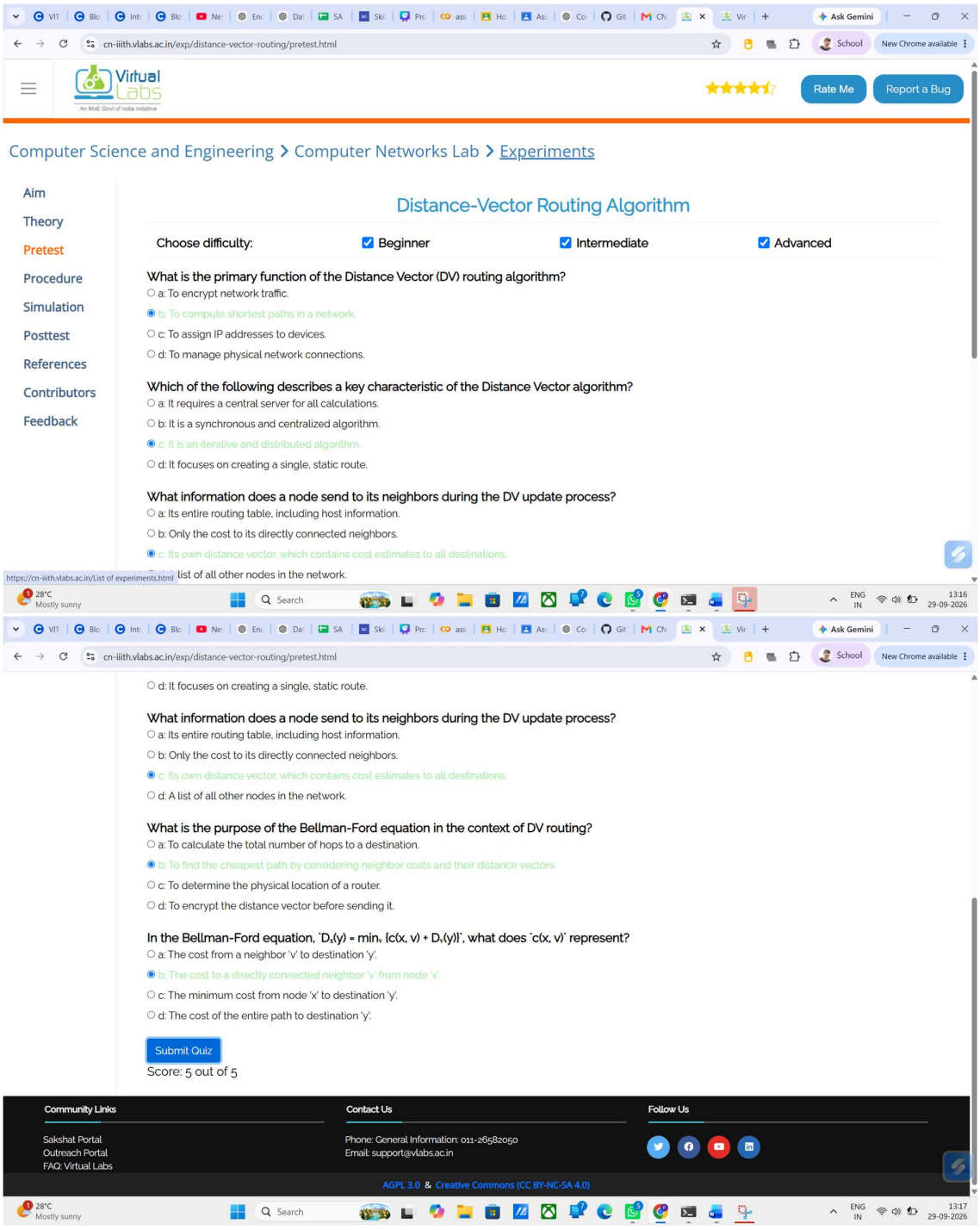


Posttest-:

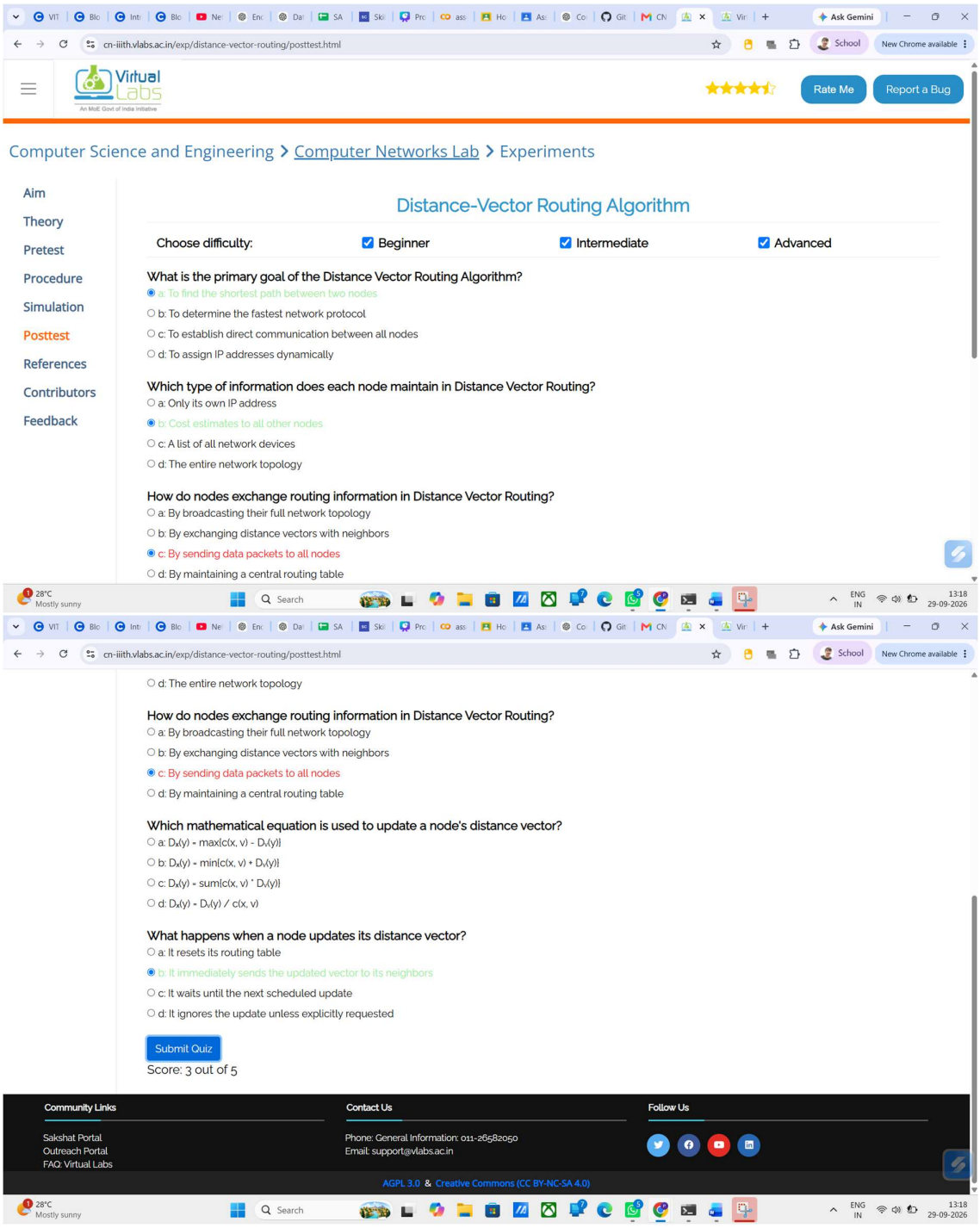

cn-iiith.vlabs.ac.in/exp/subnetting/posttest.html
Virtual Labs
An MoE Govt of India Initiative
Rate Me
Report a Bug
Computer Science and Engineering > Computer Networks Lab > Experiments
Aim
Theory
Pretest
Procedure
Simulation
Posttest
References
Contributors
Feedback
Subnetting
Choose difficulty:
Beginner
Intermediate
Advanced
What is the binary representation of the decimal number 5 using 3 bits?
a: 100
b: 101
c: 011
d: 110
If you need to create a subnet for 10 hosts, how many bits would you need for the host ID?
a: 2 bits
b: 3 bits
c: 4 bits
d: 5 bits
A network needs to be divided into 4 subnets. How many bits are required for the Subnet ID?
a: 1
b: 2
c: 3
d: 4
A network needs to be divided into 4 subnets. How many bits are required for the Subnet ID?
a: 1
b: 2
c: 3
d: 4
In a network with a subnet mask of 255.255.255.240, what is the maximum number of subnets that can be created if the original network had a /24 CIDR prefix?
a: 8
b: 16
c: 12
d: 4
You are given the network address 192.168.1.0/26. What is the Subnet ID for the third subnet?
a: 192.168.1.64
b: 192.168.1.128
c: 192.168.1.192
d: 192.168.1.32
Submit Quiz
Score: 4 out of 5
Community Links
Sakshat Portal
Outreach Portal
FAQ: Virtual Labs
Contact Us
Phone: General Information: 011-26582050
Email: support@vlabs.ac.in
Follow Us
AGPL 3.0 & Creative Commons (CC BY-NC-SA 4.0)

Title-: <u>Distance Vector Routing Algorithm</u>

Pretest-:

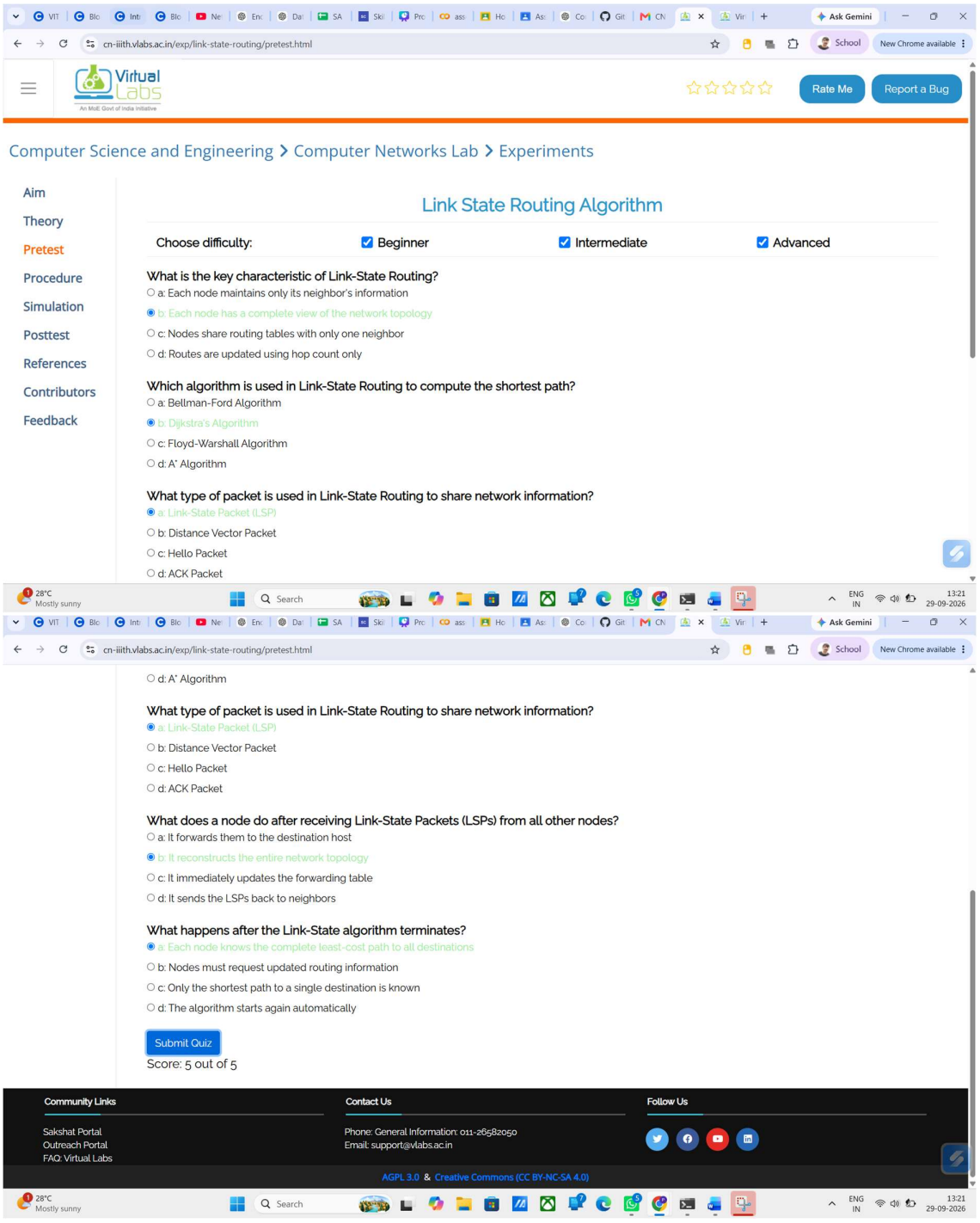


Posttest-:

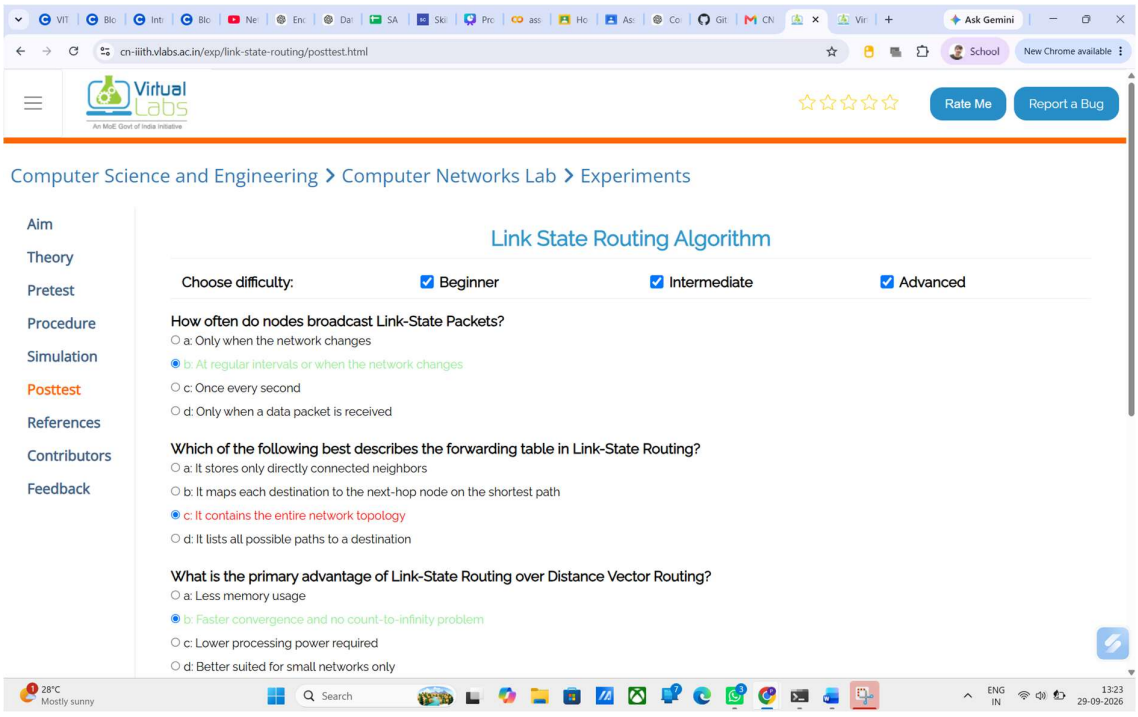

cn-iiith.vlabs.ac.in/exp/distance-vector-routing/posttest.html
Virtual Labs
An MoE Govt of India Initiative
Rate Me
Report a Bug
Computer Science and Engineering > Computer Networks Lab > Experiments
Aim
Theory
Pretest
Procedure
Simulation
Posttest
References
Contributors
Feedback
Distance-Vector Routing Algorithm
Choose difficulty:
Beginner
Intermediate
Advanced
What is the primary goal of the Distance Vector Routing Algorithm?
a: To find the shortest path between two nodes
b: To determine the fastest network protocol
c: To establish direct communication between all nodes
d: To assign IP addresses dynamically
Which type of information does each node maintain in Distance Vector Routing?
a: Only its own IP address
b: Cost estimates to all other nodes
c: A list of all network devices
d: The entire network topology
How do nodes exchange routing information in Distance Vector Routing?
a: By broadcasting their full network topology
b: By exchanging distance vectors with neighbors
c: By sending data packets to all nodes
d: By maintaining a central routing table
Which mathematical equation is used to update a node's distance vector?
a: Dx(y) = max{c(x, v) - Dv(y)}
b: Dx(y) = min{c(x, v) + Dv(y)}
c: Dx(y) = sum{c(x, v) * Dv(y)}
d: Dx(y) = Dv(y) / c(x, v)
What happens when a node updates its distance vector?
a: It resets its routing table
b: It immediately sends the updated vector to its neighbors
c: It waits until the next scheduled update
d: It ignores the update unless explicitly requested
Submit Quiz
Score: 3 out of 5
Community Links
Sakshat Portal
Outreach Portal
FAQ: Virtual Labs
Contact Us
Phone: General Information: 011-26582050
Email: support@vlabs.ac.in
Follow Us
AGPL 3.0 & Creative Commons (CC BY-NC-SA 4.0)

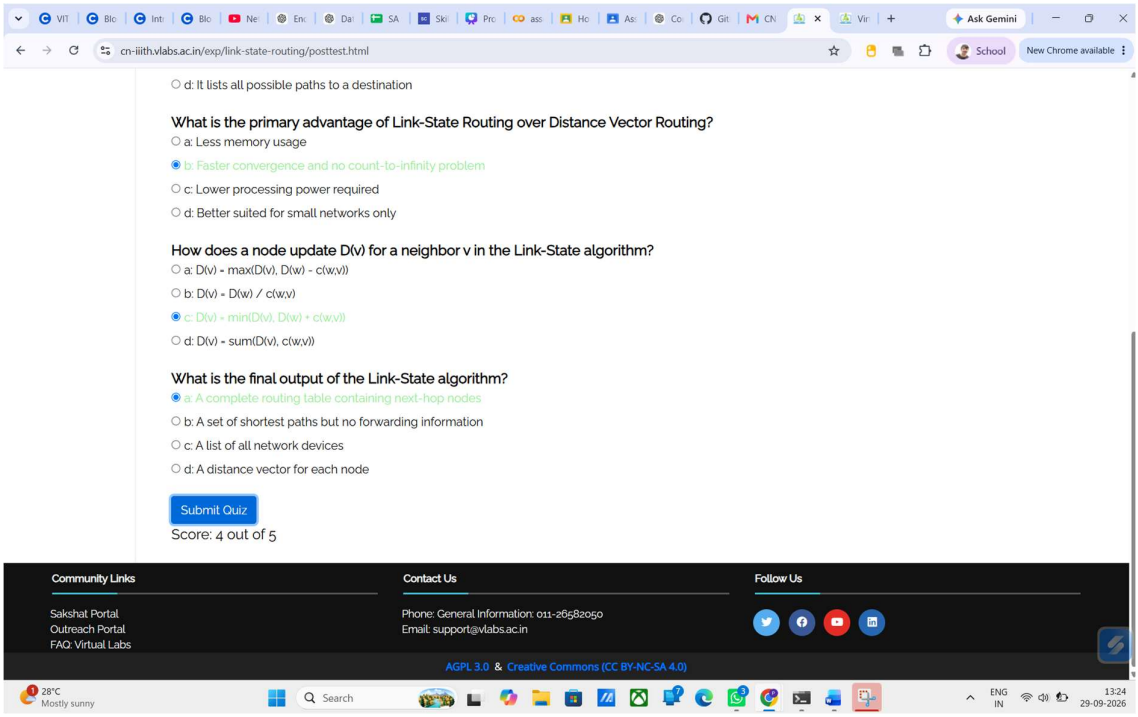
Title-: Link State Routing Algorithm

Pretest-:



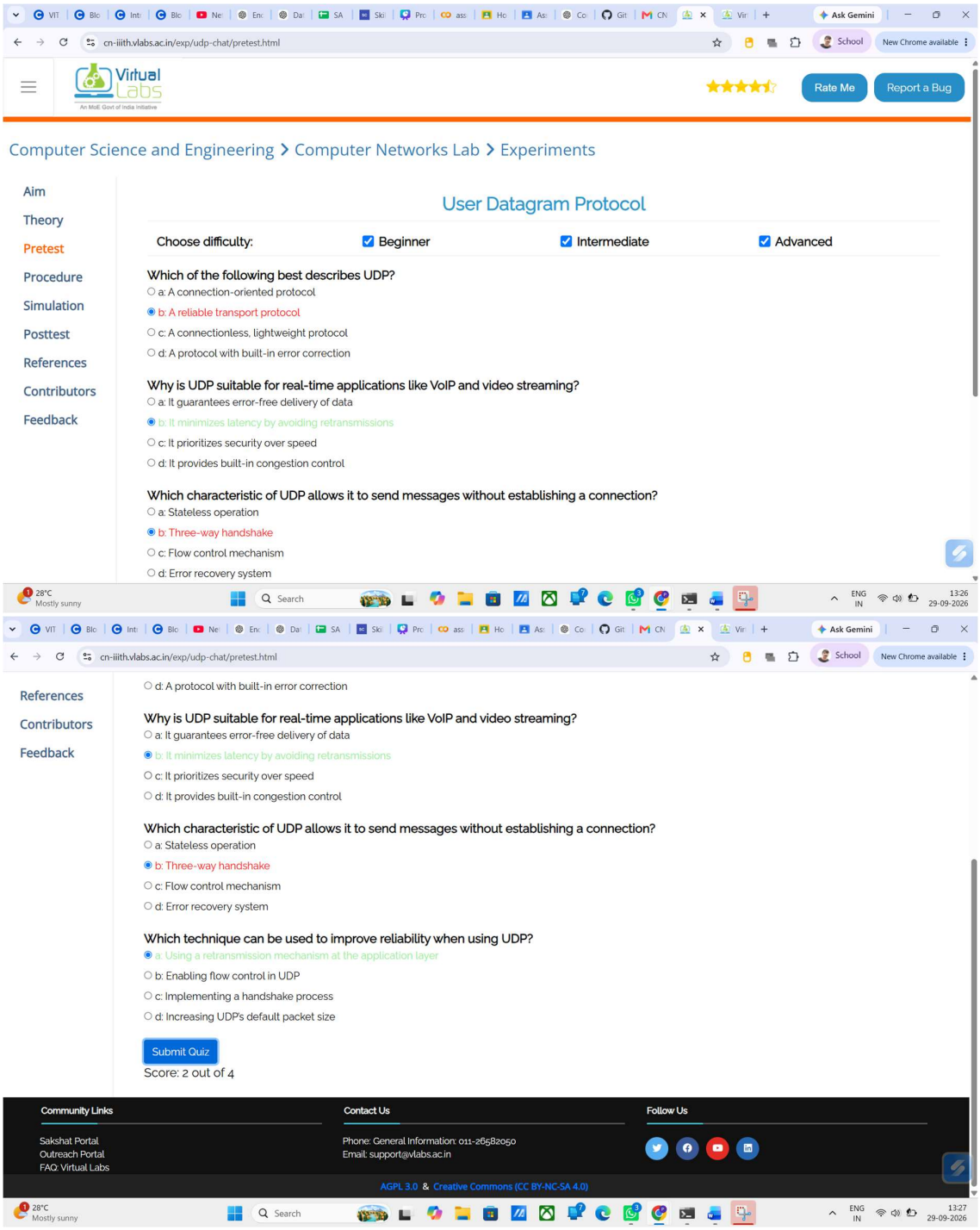
Posttest-:


cn-iiith.vlabs.ac.in/exp/link-state-routing/posttest.html
Virtual Labs
An MoE Govt of India Initiative
Rate Me
Report a Bug
Computer Science and Engineering > Computer Networks Lab > Experiments
Aim
Theory
Pretest
Procedure
Simulation
Posttest
References
Contributors
Feedback
Link State Routing Algorithm
Choose difficulty:
Beginner
Intermediate
Advanced
How often do nodes broadcast Link-State Packets?
a: Only when the network changes
b: At regular intervals or when the network changes
c: Once every second
d: Only when a data packet is received
Which of the following best describes the forwarding table in Link-State Routing?
a: It stores only directly connected neighbors
b: It maps each destination to the next-hop node on the shortest path
c: It contains the entire network topology
d: It lists all possible paths to a destination
What is the primary advantage of Link-State Routing over Distance Vector Routing?
a: Less memory usage
b: Faster convergence and no count-to-infinity problem
c: Lower processing power required
d: Better suited for small networks only

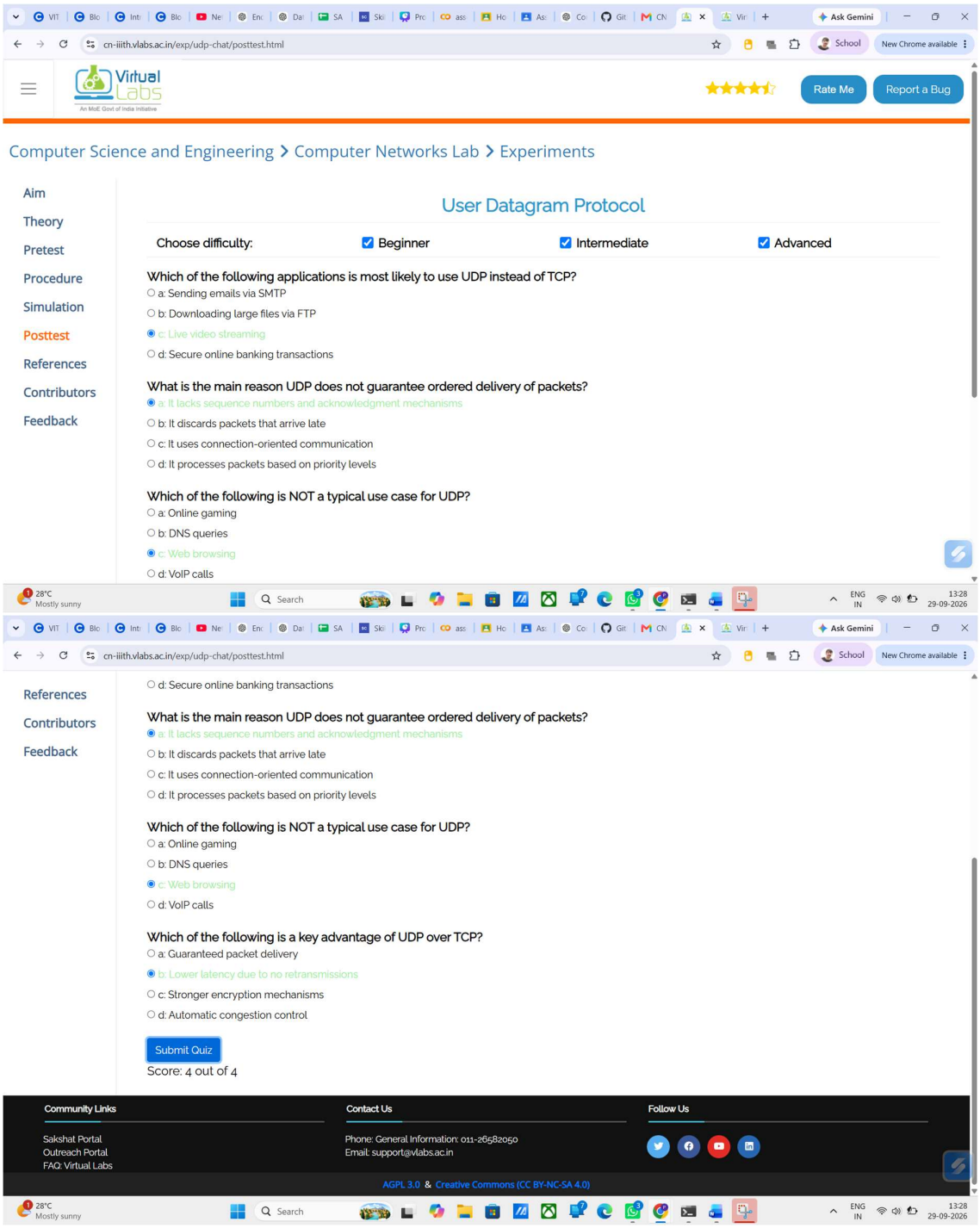

cn-iiith.vlabs.ac.in/exp/link-state-routing/posttest.html
d: It lists all possible paths to a destination
What is the primary advantage of Link-State Routing over Distance Vector Routing?
a: Less memory usage
b: Faster convergence and no count-to-infinity problem
c: Lower processing power required
d: Better suited for small networks only
How does a node update D(v) for a neighbor v in the Link-State algorithm?
a: D(v) = max(D(v), D(w) - c(w,v))
b: D(v) = D(w) / c(w,v)
c: D(v) = min(D(v), D(w) + c(w,v))
d: D(v) = sum(D(v), c(w,v))
What is the final output of the Link-State algorithm?
a: A complete routing table containing next-hop nodes
b: A set of shortest paths but no forwarding information
c: A list of all network devices
d: A distance vector for each node
Submit Quiz
Score: 4 out of 5
Community Links
Sakshat Portal
Outreach Portal
FAQ: Virtual Labs
Contact Us
Phone: General Information: 011-26582050
Email: support@vlabs.ac.in
Follow Us
AGPL 3.0 & Creative Commons (CC BY-NC-SA 4.0)

Title-: Chat Application using UDP

Pretest-:

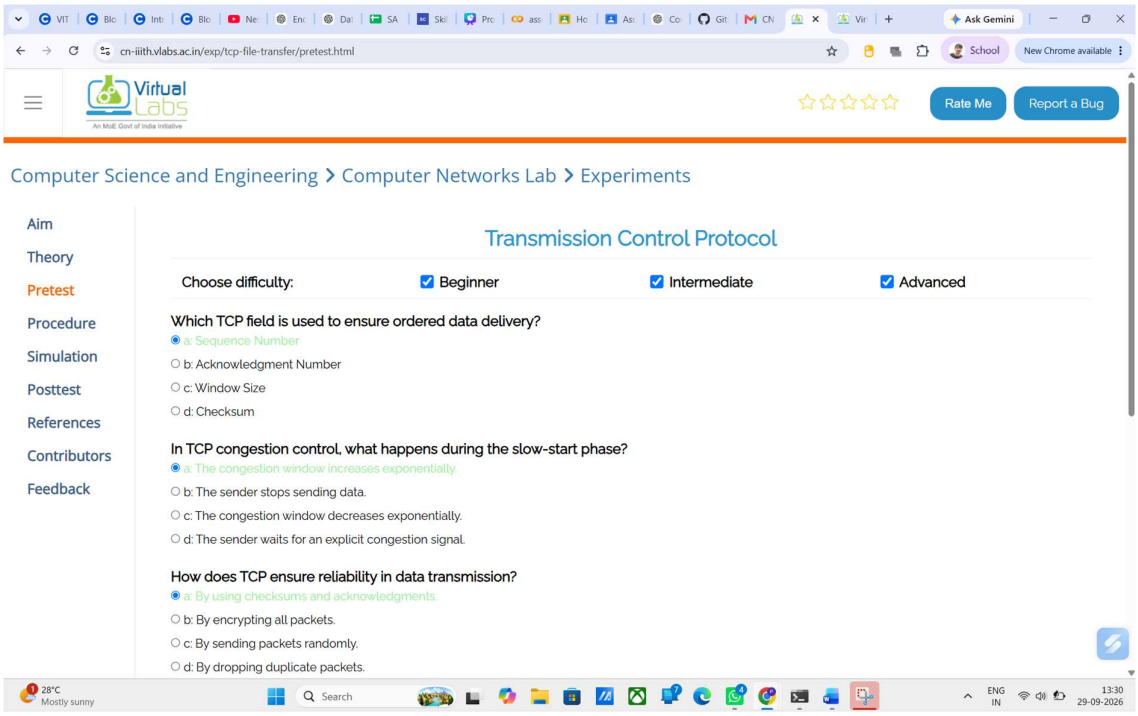


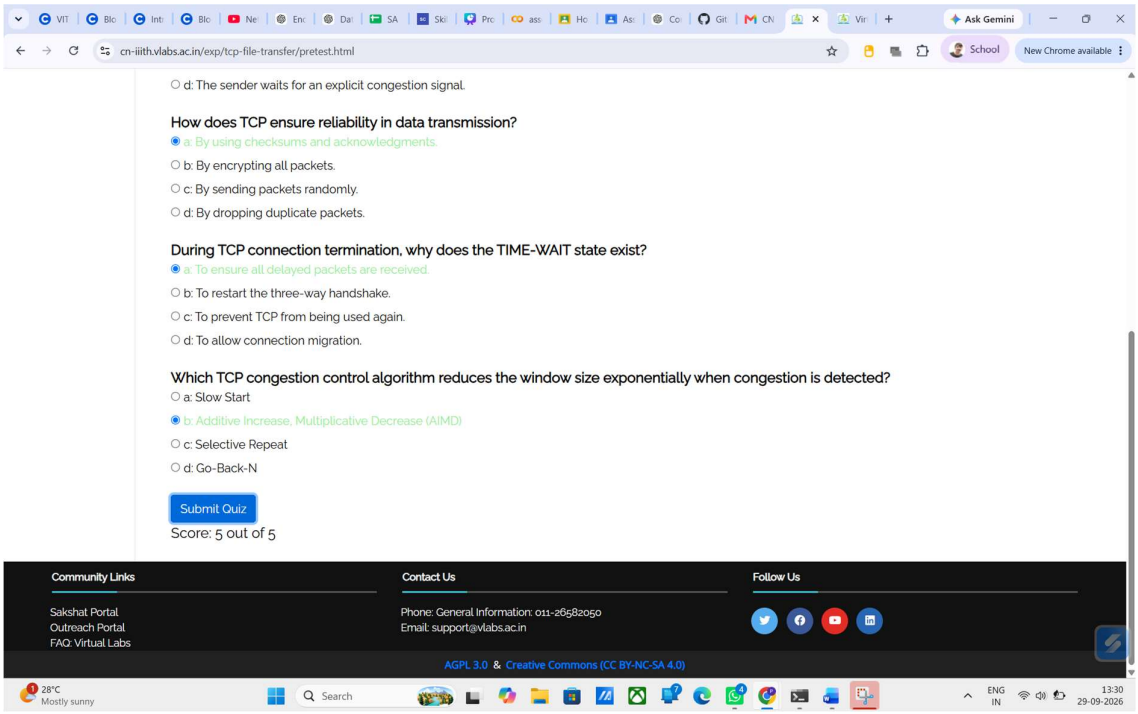
Posttest-:


cn-iiith.vlabs.ac.in/exp/udp-chat/posttest.html
Virtual Labs
An MoE Govt of India Initiative
Rate Me
Report a Bug
Computer Science and Engineering > Computer Networks Lab > Experiments
Aim
Theory
Pretest
Procedure
Simulation
Posttest
References
Contributors
Feedback
User Datagram Protocol
Choose difficulty:
Beginner
Intermediate
Advanced
Which of the following applications is most likely to use UDP instead of TCP?
a: Sending emails via SMTP
b: Downloading large files via FTP
c: Live video streaming
d: Secure online banking transactions
What is the main reason UDP does not guarantee ordered delivery of packets?
a: It lacks sequence numbers and acknowledgment mechanisms
b: It discards packets that arrive late
c: It uses connection-oriented communication
d: It processes packets based on priority levels
Which of the following is NOT a typical use case for UDP?
a: Online gaming
b: DNS queries
c: Web browsing
d: VoIP calls
Which of the following is a key advantage of UDP over TCP?
a: Guaranteed packet delivery
b: Lower latency due to no retransmissions
c: Stronger encryption mechanisms
d: Automatic congestion control
Submit Quiz
Score: 4 out of 4
Community Links
Sakshat Portal
Outreach Portal
FAQ: Virtual Labs
Contact Us
Phone: General Information: 011-26582050
Email: support@vlabs.ac.in
Follow Us
AGPL 3.0 & Creative Commons (CC BY-NC-SA 4.0)

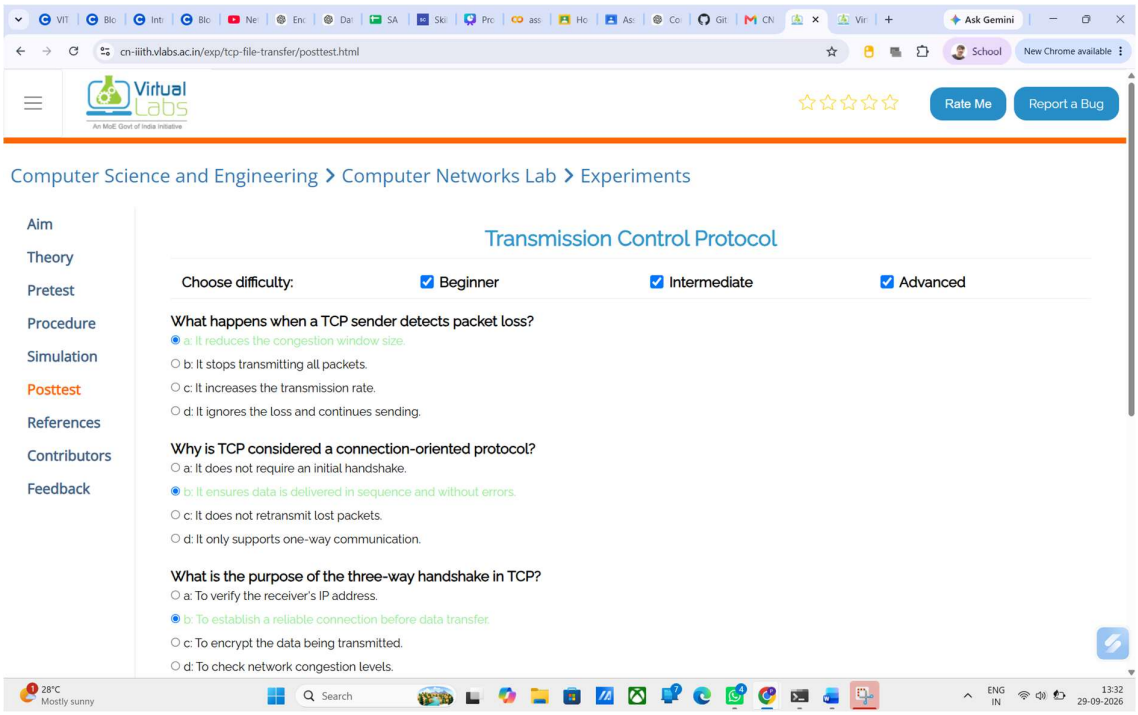
Title-: File Transfer using TCP

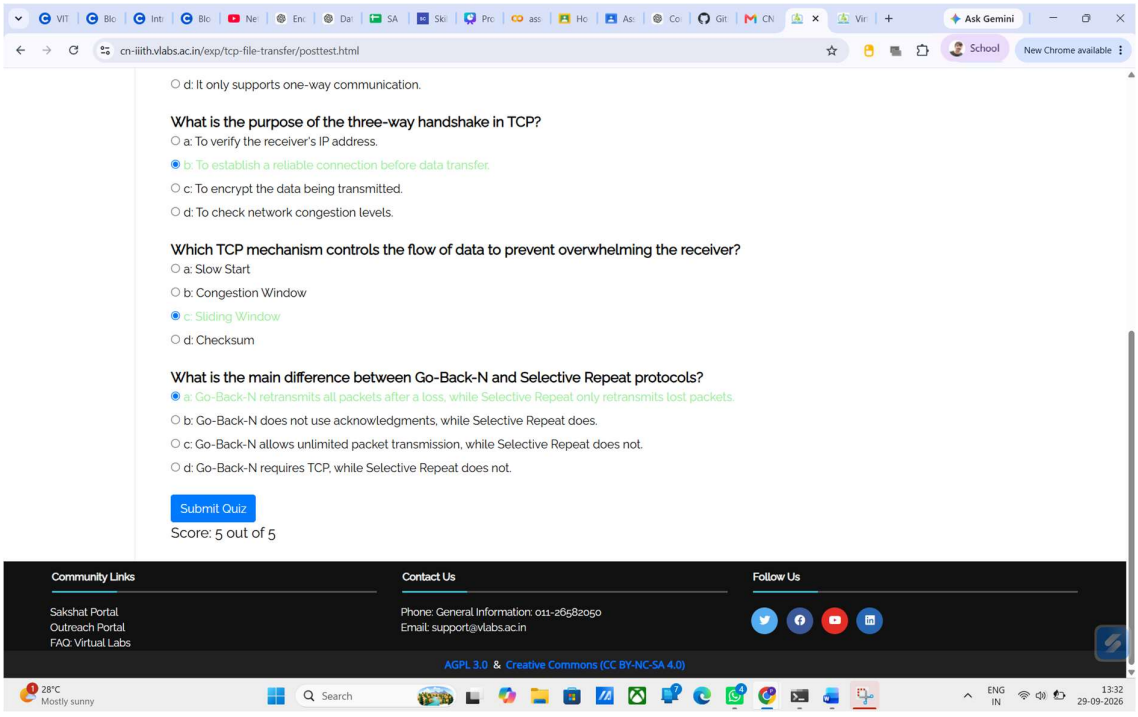
Pretest-:





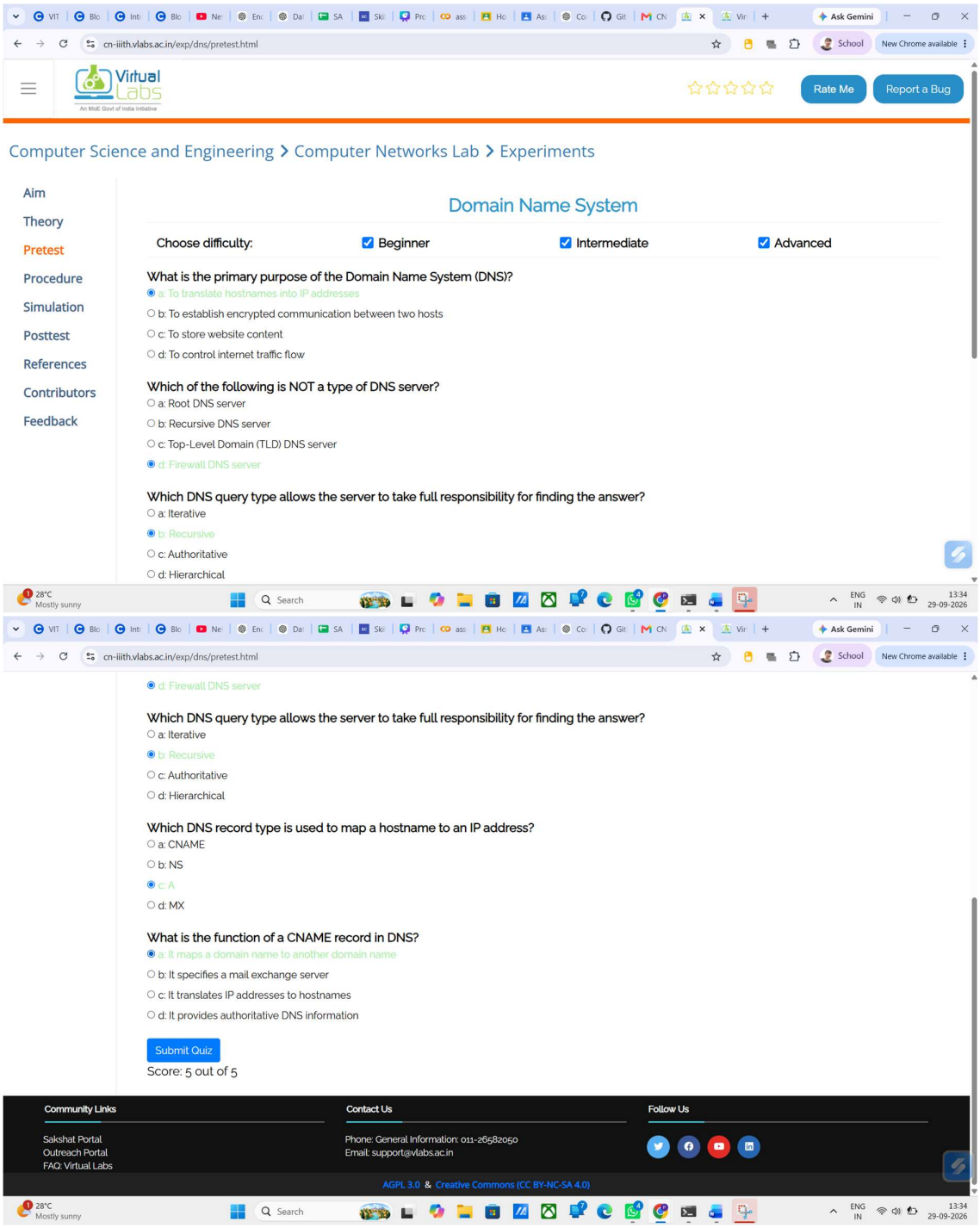
Posttest-:


cn-iiith.vlabs.ac.in/exp/tcp-file-transfer/posttest.html
Virtual Labs
An MoE Govt of India Initiative
Rate Me
Report a Bug
Computer Science and Engineering > Computer Networks Lab > Experiments
Aim
Theory
Pretest
Procedure
Simulation
Posttest
References
Contributors
Feedback
Transmission Control Protocol
Choose difficulty:
Beginner
Intermediate
Advanced
What happens when a TCP sender detects packet loss?
a: It reduces the congestion window size.
b: It stops transmitting all packets.
c: It increases the transmission rate.
d: It ignores the loss and continues sending.
Why is TCP considered a connection-oriented protocol?
a: It does not require an initial handshake.
b: It ensures data is delivered in sequence and without errors.
c: It does not retransmit lost packets.
d: It only supports one-way communication.
What is the purpose of the three-way handshake in TCP?
a: To verify the receiver's IP address.
b: To establish a reliable connection before data transfer.
c: To encrypt the data being transmitted.
d: To check network congestion levels.

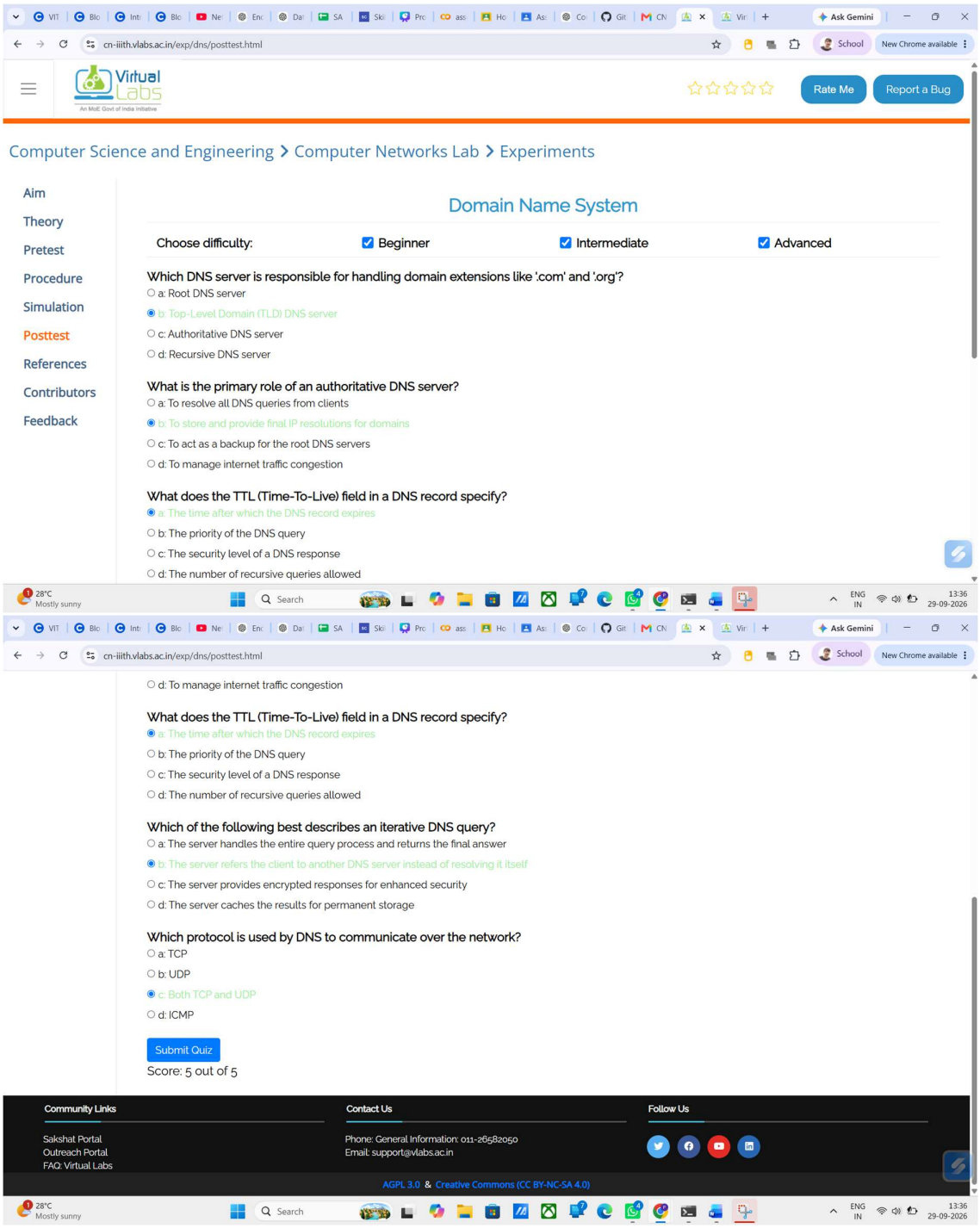

cn-iiith.vlabs.ac.in/exp/tcp-file-transfer/posttest.html
d: It only supports one-way communication.
What is the purpose of the three-way handshake in TCP?
a: To verify the receiver's IP address.
b: To establish a reliable connection before data transfer.
c: To encrypt the data being transmitted.
d: To check network congestion levels.
Which TCP mechanism controls the flow of data to prevent overwhelming the receiver?
a: Slow Start
b: Congestion Window
c: Sliding Window
d: Checksum
What is the main difference between Go-Back-N and Selective Repeat protocols?
a: Go-Back-N retransmits all packets after a loss, while Selective Repeat only retransmits lost packets.
b: Go-Back-N does not use acknowledgments, while Selective Repeat does.
c: Go-Back-N allows unlimited packet transmission, while Selective Repeat does not.
d: Go-Back-N requires TCP, while Selective Repeat does not.
Submit Quiz
Score: 5 out of 5
Community Links
Sakshat Portal
Outreach Portal
FAQ: Virtual Labs
Contact Us
Phone: General Information: 011-26582050
Email: support@vlabs.ac.in
Follow Us
AGPL 3.0 & Creative Commons (CC BY-NC-SA 4.0)

Title-: Domain Name System (DNS)

Pretest-:



Posttest-:


cn-iiith.vlabs.ac.in/exp/dns/posttest.html
Virtual Labs
An MoE Govt of India Initiative
Rate Me
Report a Bug
Computer Science and Engineering > Computer Networks Lab > Experiments
Aim
Theory
Pretest
Procedure
Simulation
Posttest
References
Contributors
Feedback
Domain Name System
Choose difficulty:
Beginner
Intermediate
Advanced
Which DNS server is responsible for handling domain extensions like '.com' and '.org'?
a: Root DNS server
b: Top-Level Domain (TLD) DNS server
c: Authoritative DNS server
d: Recursive DNS server
What is the primary role of an authoritative DNS server?
a: To resolve all DNS queries from clients
b: To store and provide final IP resolutions for domains
c: To act as a backup for the root DNS servers
d: To manage internet traffic congestion
What does the TTL (Time-To-Live) field in a DNS record specify?
a: The time after which the DNS record expires
b: The priority of the DNS query
c: The security level of a DNS response
d: The number of recursive queries allowed
Which of the following best describes an iterative DNS query?
a: The server handles the entire query process and returns the final answer
b: The server refers the client to another DNS server instead of resolving it itself
c: The server provides encrypted responses for enhanced security
d: The server caches the results for permanent storage
Which protocol is used by DNS to communicate over the network?
a: TCP
b: UDP
c: Both TCP and UDP
d: ICMP
Submit Quiz
Score: 5 out of 5
Community Links
Sakshat Portal
Outreach Portal
FAQ: Virtual Labs
Contact Us
Phone: General Information: 011-26582050
Email: support@vlabs.ac.in
Follow Us
AGPL 3.0 & Creative Commons (CC BY-NC-SA 4.0)